

SZOMBATHELY AP VAS, Hungary

WMO: 128120

Lat: 47.1983N

Lon: 16.6478E

Elev: 725

StdP: 14.31

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.8	(c) 17.5	(d) 5.6	(e) 7.5	(f) 15.4	(g) 10.6	(h) 9.7	(i) 20.3	(j) 31.7	(k) 30.7	(l) 27.0	(m) 31.6	(n) 5.6	(o) 10	(p) 0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.2	(c) 90.0	(d) 69.5	(e) 86.8	(f) 68.9	(g) 83.6	(h) 67.6	(i) 71.8	(j) 84.4	(k) 70.2	(l) 83.1	(m) 68.8	(n) 81.0	(o) 7.5	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 67.7	(b) 104.4	(c) 76.9	(d) 66.0	(e) 98.6	(f) 75.0	(g) 64.6	(h) 93.6	(i) 73.6	(j) 35.9	(k) 84.3	(l) 34.5	(m) 83.1	(n) 33.3	(o) 81.1	(p) 78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.3	(b) 18.3	(c) 15.4	(e) 6.1	(f) 94.7	(g) 6.0	(h) 3.6	(i) 1.8	(j) 97.3	(k) -1.8	(l) 99.4	(m) -5.1	(n) 101.4	(o) -9.5	(p) 104.1		
(a) 22.3	(b) 18.3	(c) 15.4	(e) 5.6	(f) 74.6	(g) 5.8	(h) 2.1	(i) 1.4	(j) 76.1	(k) -2.0	(l) 77.3	(m) -5.3	(n) 78.4	(o) -9.5	(p) 79.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	31.6	35.1	42.6	51.6	59.6	66.4	69.7	69.0	60.5	51.4	42.4	33.4	
	DBStd	15.02	7.47	8.13	7.82	6.79	5.82	6.10	5.50	5.52	5.74	6.74	7.31	7.14	
	HDD50	2126	570	420	246	61	3	0	0	0	1	66	243	516	
	HDD65	5490	1035	838	694	402	184	56	21	23	157	421	678	981	
	CDD50	2564	0	1	17	110	300	492	612	589	317	110	15	1	
	CDD65	454	0	0	0	1	16	99	168	146	23	1	0	0	
	CDH74	4761	0	0	0	30	249	1018	1725	1499	228	12	0	0	
	CDH80	1576	0	0	0	2	37	304	627	567	39	0	0	0	
Wind	WSAvg	6.2	5.9	6.7	7.2	7.5	6.7	6.3	6.1	5.4	5.7	5.2	5.7	6.0	
Precipitation	PrecAvg	25.6	1.1	1.1	1.4	1.8	2.6	3.2	3.1	3.0	2.6	2.2	2.1	1.5	
	PrecMax	35.0	2.6	3.5	3.0	3.3	4.9	6.2	5.7	6.3	6.1	5.2	5.2	3.0	
	PrecMin	18.6	0.3	0.2	0.1	0.0	0.9	0.8	0.2	0.4	0.8	0.1	0.2	0.2	
	PrecStd	4.0	0.6	0.9	0.8	0.8	1.1	1.3	1.5	1.5	1.5	1.3	1.1	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	61.6	68.7	78.1	83.8	90.8	94.5	94.9	84.4	75.6	65.5	55.6	
		MCWB	46.5	49.3	52.3	60.8	65.2	69.6	70.6	69.0	67.2	62.3	56.6	50.0	
	2%	DB	49.0	55.5	64.4	73.1	79.9	87.1	89.8	90.3	79.7	70.8	60.1	49.2	
		MCWB	43.7	46.7	51.4	57.5	64.3	69.6	69.6	69.3	65.0	60.7	53.8	45.2	
	5%	DB	44.9	51.1	60.2	69.3	76.6	83.6	86.5	86.4	76.2	66.8	56.0	45.5	
		MCWB	40.9	44.3	49.6	55.7	62.7	68.2	68.7	68.8	64.0	58.7	51.8	41.9	
	10%	DB	41.4	46.9	55.8	65.8	72.9	80.2	83.3	82.4	72.5	62.9	52.7	42.5	
		MCWB	38.6	41.5	47.2	53.6	60.5	66.3	67.5	67.2	61.8	56.6	49.2	40.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	50.1	54.5	61.9	68.3	73.4	73.8	73.7	69.1	64.1	57.7	50.4
			MCDB	53.2	59.8	64.9	75.6	79.9	85.4	87.2	85.6	81.0	73.0	64.1	54.7
2%		WB	44.3	47.3	52.2	58.8	65.7	71.0	71.5	71.6	67.0	61.4	54.5	45.6	
		MCDB	48.1	54.6	62.1	70.7	77.0	82.9	85.0	84.2	76.9	68.9	59.1	48.6	
5%		WB	41.4	44.7	50.1	56.6	63.7	69.2	70.0	69.9	65.1	59.4	52.2	42.5	
		MCDB	44.3	50.4	58.8	66.8	74.2	80.9	83.2	83.0	73.3	65.6	55.9	44.5	
10%		WB	39.0	42.0	48.0	54.6	61.7	67.4	68.6	68.2	63.0	57.5	49.6	40.4	
		MCDB	41.0	46.4	54.8	64.2	71.1	78.3	80.7	80.4	70.6	62.3	52.1	42.2	
Mean Daily Temperature Range		5% DB	MDBR	10.9	14.2	17.6	20.2	19.8	20.6	21.2	21.2	18.5	17.6	11.9	9.9
			MCDBR	16.5	22.1	26.0	28.1	27.0	26.4	27.7	28.4	24.7	23.3	18.5	14.9
	MCWBR		11.8	14.3	15.1	14.6	13.3	11.1	10.1	10.4	11.8	13.7	12.9	11.2	
	5% WB	MCDBR	15.6	20.5	23.7	24.9	24.3	24.5	24.8	25.5	22.0	21.5	17.5	14.0	
		MCWBR	11.5	13.7	14.3	13.4	12.5	11.1	10.1	10.8	11.2	13.1	12.4	11.0	
Clear-Sky Solar Irradiance	taub	0.320	0.344	0.381	0.426	0.422	0.427	0.431	0.434	0.409	0.380	0.350	0.320		
	taud	2.390	2.343	2.268	2.160	2.222	2.248	2.237	2.241	2.283	2.338	2.375	2.414		
	Ebn at Noon	240	258	264	262	268	267	263	257	252	240	225	226		
	Edh at Noon	23	29	36	44	43	42	42	41	36	29	23	20		
All-Sky Solar Radiation	RadAvg	353	609	993	1439	1726	1879	1845	1637	1164	746	396	270		
	RadStd	38	86	120	139	165	131	139	162	137	76	41	35		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (85 neighbors)	+1.03	N/A	N/A	+1.62	+0.88	+0.79	-209	-288	+175	+103

Nomenclature: See separate page